STATISTICS · ANTICIPATING PATTERNS

Conditional Probability: Screening Tests

Explore & Reflect TEACHER KEY

Name _____ Date _____ Period _____

Open **Conditional Probability: Screening Tests** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A test is 99% sensitive and 99% specific, and 1% of people have the condition. If you test positive, guess the chance that you have it.
   It is only 50%. (Most people guess about 99%.)
2. Is P(positive | condition) the same as P(condition | positive)? Explain.
   No. They divide by different groups: everyone with the condition vs. everyone who tests positive.

2 Explore use the tool

1. Set prevalence 1%, sensitivity 99%, specificity 99%. Record the true positives, false positives and P(C | +).
   990 true positives, 990 false positives (out of 100,000); P(C | +) = 50%.
2. Keep the 99%/99% test. Which prevalence on the slider is the first to make P(C | +) at least 90%?
   10% (P(C | +) ≈ 91.7%; 5% gives ≈ 83.9%).
3. Prevalence 5%: compare raising sensitivity to 100% with raising specificity to 99.5%. Which helps P(C | +) more? Why?
   Specificity: 99.5% gives ≈ 91.2% but 100% sensitivity only ≈ 84.0%. Most positives come from the large no-condition group, so fewer false positives matters most.
4. Set sensitivity 50% and specificity 50%. What is P(C | +)? Why?
   It equals the prevalence: the result is independent of the condition, so a positive tells you nothing.

3 Record use the tool and fill in the table

P(condition | positive) and P(condition | negative)

Prevalence	Sensitivity	Specificity	P(C | +)	P(C | −)
0.1%	99%	99%	≈ 9.0%	≈ 0.001%
2%	90%	95%	≈ 26.9%	≈ 0.21%
20%	90%	95%	≈ 81.8%	≈ 2.56%
50%	80%	60%	≈ 66.7%	25%

4 Explain what did you discover?

1. P(C | +) = true positives ÷ all positives .
2. When a condition is rare, many positive results are false positives .
3. Events A and B are independent when P(A | B) = P(A) (and P(A | not B)) .

4. Explain to a patient why a positive result on a 99%-accurate test (99% sensitive and specific) for a disease that affects 0.1% of people is probably a false positive. Use counts out of 100,000.
   Of 100,000 people, 100 have the disease and 99 test positive. Of the 99,900 who don't, 1% (999) also test positive. Only 99 of the 1,098 positives (about 9%) have the disease.

5 Apply a new situation, no tool

1. At a school, 45 of 60 seniors and 30 of 90 juniors drive to school. Find P(drives | senior) and P(senior | drives). Are "senior" and "drives" independent?
   0.75 and $\frac{45}{75}$ = 0.6. P(drives) = $\frac{75}{150}$ = 0.5 ≠ 0.75, so not independent.
2. P(A) = 0.3, P(B | A) = 0.4 and P(B | not A) = 0.2. Find P(A | B).
   $\frac{0.12}{(0.12 + 0.14)}$ ≈ 0.4615
3. 40% of email is spam. A filter flags 95% of spam and 2% of other email. Find P(spam | flagged).
   $\frac{0.38}{(0.38 + 0.012)}$ ≈ 0.969

Teacher notes

Intended discovery: P(A | B) and P(B | A) have different denominators; for rare conditions false positives can outnumber true positives, so P(condition | positive) depends heavily on prevalence and specificity; independence means P(+ | C) = P(+ | not C).

Likely misconceptions:

- Swapping P(A | B) and P(B | A) (the prosecutor's fallacy).
- Ignoring the base rate.
- Thinking a highly sensitive test makes a positive result nearly certain.

Answers that can vary: Predict 1 is a guess; the key gives the correct value.